Secondary Chilled Water Pumps #1, #2 and #3 Sequence:

The Secondary Chilled Water Pumps (SCHWP) shall be indexed to run, as needed, any time the chiller plant is manually started.

The SCHWPs shall maintain equal run times during the year. They will be alternated every Sunday:

- P1, P2, P3
- P2, P3, P1
- P3, P1, P2

Chilled Water Pump Control

a. Chilled water pump hand-off-auto operation: HOA settings shall be provided as part of the variable frequency drive through the drive's keypad. In the off mode, the pump shall be stopped. In the hand mode, the pump shall run continuously. In the auto mode, the pump shall be started and stopped by the BAS.
b. Chilled water pump local-remote speed control: local-remote settings shall be provided as part of each variable frequency drive through the drive's keypad. In the local mode, the pump's speed shall be controlled through a manual speed control located at the respective drive control panel. In the remote mode, the pump's speed shall be controlled by the BAS.
c. When the chilled water system is enabled, the lead chiller's primary pump is commanded to run. Once the lead chilled water primary pump is enabled and runs at minimum speed (30% adj.), it is then placed under control of its variable frequency drive. The minimum speed shall be hard programmed into the VFD. The minimum speed shall be determined by the water balancer as the minimum speed required to provide minimum flow to the chiller.
d. The BAS modulates the speed of the secondary chilled water pumps via its VFD to maintain the system chilled water distribution system differential pressure at the system extremities. A differential pressure sensor is provided in the piping vault serving building 207. The BAS shall poll the aforementioned sensor and the pumps' VFD shall be controlled to satisfy the lowest differential pressure set point.
e. The balancing agent shall provide minimum and design flow pressure settings for the differential pressure sensor.
f. The BAS uses pump current sensors and VFD power output to confirm the pump is in the desired state (on or off). If the lead secondary chilled water pump fails to start, the pump is stopped and an alarm is initiated through the BAS. The next available (lag) pump is started.
g. Additional chilled water pumps are started such that the number of pumps running satisfies the differential pressure set point. As additional pumps are started they shall run at the same speed to maintain the differential pressure at set point.

When the lead pump is indexed to run, it shall ramp up to maintain the differential pressure setpoint.

When the differential pressure PID is greater than 80% output, the second SCHWP will be indexed to run and both pumps will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is greater than 80% again, the third SCHWP will be indexed to run and all three pumps will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is less than 20%, the third SCHWP started will be indexed to stop and the remaining two SCHWPs will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is less than 20% output again, the second SCHWP will be indexed to stop and the lead SCHWP will be controlled to maintain the differential pressure setpoint.

An alarm will be activated if the differential pressure is above or below setpoint by more than 0.4 psi (adj.) for more than 2 min, 30 seconds (adj.).

An alarm will be activated if any of the three pumps speed feedback is 5% (adj.) above or below commanded speed for more than 30 seconds.

AHU CHW requests used for pump DP optimization reset:

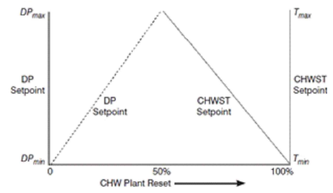
The central chilled water plant shall utilize a pump differential pressure optimization and chilled water temperature reset strategy. Air handling units and cooling control valves shall be programmed to send position and cooling setpoint deviation signals to the BAS. The BAS shall be capable of assigning priority based on deviation from design supply air setpoint. 1°F deviation from setpoint will set first level of priority and initiate a call for differential pressure reset. Upon a continued deviation up to 2°F from setpoint the BAS shall send an alarm notification. Refer to chilled water plant sequences listed hereinafter for coordination.

Chilled Water Pump Speed Control - Optimized Differential Pressure Reset and Chilled Water Temperature Reset (Trim and Respond)

The objective is to minimize pump energy by always operating the pump(s) at the lowest speed possible such that at least one chilled water control valve is fully open and maintaining its temperature control set point within an acceptable range. The minimum pump speed shall be programmed at the VFD drive for (typ. 20% or 12 hz adj.) and also in the software as necessary. The maximum pump speed shall be programmed at the VFD drive for (typ. 100% or 60 Hz) and also in the software as necessary.

The diagram below shows how this sequenced reset strategy shall be implemented. The x-axis is a software point called “CHW Plant Reset” that varies from 0% to 100% using “trim and respond” logic. In general, the AHU coil valve controllers generate “requests” for colder chilled water temperature or higher pump pressure when the valve is full open. When valves are generating “requests,” CHW Plant Reset increases; when they are not, CHW Plant Reset steadily decreases.

When CHW Plant Reset is 100%, the CHWST setpoint is at Tmin (the design chilled water temperature of 42°F) and the DP setpoint is at DPmax (the design DP setpoint). As the load backs off and the cooling requests from the AHUs decrease, the trim and respond logic reduces the CHW Plant Reset point. As it does, chilled water temperature is increased first up to a maximum Tmax (equal to the lowest air handler supply air temperature setpoint less 2°F) or 48F (adj.), then DP setpoint is reduced down to a minimum value DPmin (such as 5 psi).



Once the pump is enabled the pump will start at minimum speed. The optimized setpoint block (CHW Plant Reset) shall modulate the pump VFD to its maximum speed, as required to maintain a differential pressure setpoint. Initially, the chilled water plant shall generate chilled water at Tmin (design chilled water temperature of 42°F).

The differential pressure setpoint and chilled water setpoint shall be automatically reset with a trim and respond function, when commanded by the BAS as described below.

a. The chilled water supply temperature setpoint shall be recalculated every 5 minutes (adj.). The incremental change per interval shall be based on the maximum cooling requests from any one of the associated primary or terminal controllers served by each pump system. A cooling request is defined as a condition where the requesting equipment's control valve is 100% open and the controlled variable of the valve is off setpoint by 1°F (e.g. 1 cooling request). Each additional 1°F from setpoint will generate an additional request. Therefore, the further from setpoint the greater the number of requests from that particular piece of equipment. The chilled water plant reset optimized setpoint block resets based on any given valve controller with the highest number of requests (requests are not added from various valve controllers).

b. The chilled water supply temperature setpoint shall be reset using the trim and respond logic described hereinafter. The initial setpoint shall be 42°F (adj). During chilled water plant reset mode, the setpoint shall be reset from Tmin 42°F (adj.), up to a Tmax of 48°F.

c. As cooling requests increase, the setpoint shall incrementally respond up by 0.25 °F per request value, limited to 1°F (adj.) per period (see above for time period), to the maximum reset value. With no cooling requests, the setpoint shall incrementally be trimmed down by 0.25 °F (adj.) per period to the minimum reset value.

d. If network communication is lost for more than 10 minutes, the setpoint shall be reset to 42°F and an alarm issued.

e. The pump VFD speed DP setpoint shall be recalculated every 5 minutes (adj.). The incremental change per interval shall be based on the maximum cooling requests from any one of the associated primary or terminal controllers served by each pump system. A cooling request is defined as a condition where the requesting equipment's control valve is 100% open and the controlled variable of the valve is off setpoint by 1°F (e.g. 1 cooling request). Each additional 1°F from setpoint will generate an additional request. Therefore, the further from setpoint the greater the number of requests from that particular equipment. The pump enable optimized setpoint block resets based on any given valve controller with the highest number of requests (requests are not added from various valve controllers).

f. The pump VFD speed differential pressure setpoint shall be reset between a min and max setpoint (5-12 for example) psi. The initial differential pressure setpoint shall be 5 psig (adj). The maximum DP setpoint is the pressure required to provide full pump design flow (taking into account the design diversity of all the connected load) to all control valves simultaneously while providing full design flow to the “hydraulically most remote” load.

g. The minimum value is an arbitrarily set number representing a low value that will likely result in pump operation at min speed for optimal energy savings but still high enough to maintain minimal flow after enable until DP has time to reset if required.). If network communication is lost for more than 10 minutes, the setpoint shall be reset to 0.75 x maximum DP setpoint and an alarm issued.

h. As cooling requests increase, the setpoint shall incrementally respond up by 0.5 psi per request value, limited to 2 psi (adj.) per period (see above for time period), to the maximum reset value. With no cooling requests, the setpoint shall incrementally be trimmed down by 0.5 psi (adj.) per period to the minimum reset value.

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1	TDP-1 COMMENT REVISIONS 2	07/06/18		APPR'd

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BUILDING 106 - SECONDARY CHW SYSTEM SEQUENCE OF OPERATION

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SHEET 11 OF 44